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Prevalence of antibiotic resistance genes in *Pangasianodon hypophthalmus* and *Oreochromis niloticus* aquaculture production systems in Bangladesh

Simon Bo Lassen^{a,c,d,1}, Md. Emranul Ahsan^{a,b,e,1}, Seikh Razibul Islam^{a,b}, Xin-Yuan Zhou^c, Muhammad Abdur Razzak^b, Jian-Qiang Su^c, Kristian Koefoed Brandt^{a,d,*}

^a Department of Plant and Environmental Sciences, University of Copenhagen, Thorvaldsensvej 40, 1871 Frederiksberg, Denmark

^b Department of Aquaculture, Patuakhali Science and Technology University, Dumki, Patuakhali 8602, Bangladesh

^c Key Laboratory of Urban Environment and Health, Institute of Urban Environment, Chinese Academy of Sciences, 1799 Jimei Road, Xiamen 361021, China

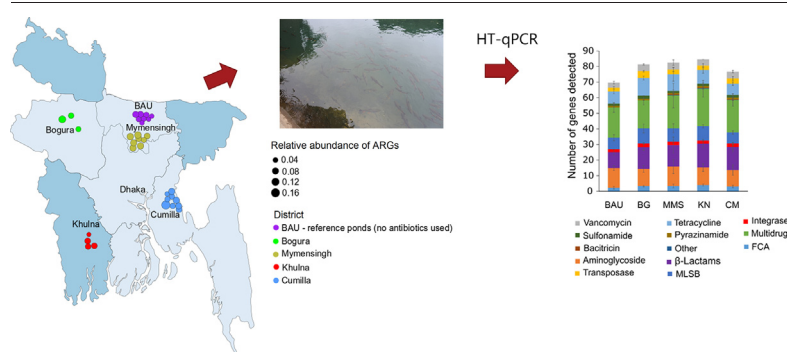
^d Sino-Danish Center for Education and Research (SDC), University of Chinese Academy of Sciences, 380 Huaibeizhuang, Beijing, China

^e Department of Fisheries Management, Bangabandhu Sheikh Mujibur Rahman Agricultural University, Salna, Gazipur 1706, Bangladesh

HIGHLIGHTS

- Antibiotic resistance genes (ARGs) were quantified in aquaculture ponds in Bangladesh.
- 160 unique ARGs and 10 MGEs were detected, with 76 genes detected across all regions.
- Ponds with no antibiotic-use and frequent water exchange had lowest ARGs abundance.
- ARG prevalence patterns could be statistically associated with geography and feed type.
- The abundance of ARGs was similar to other anthropogenically impacted waters.

GRAPHICAL ABSTRACT



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ABSTRACT

Antibiotic resistance genes (ARGs) constitute emerging pollutants of significant public health concern. Antibiotics applied in aquaculture may stimulate the proliferation and dissemination of ARGs. This study investigated the prevalence and diversity of ARGs in *Pangasianodon hypophthalmus* (formerly *Pangasius*) and *Oreochromis niloticus* (formerly *Tilapia*) commercial aquaculture ponds from four economically important divisions (i.e. regions) of Bangladesh using a high-throughput qPCR ARG SmartChip and further aimed to explore effects of aquaculture pond management and water quality on the observed ARG prevalence patterns. A total of 160 ARGs and 10 mobile genetic elements (MGEs) were detected across all samples ($n = 33$), of which 76 ARGs and MGEs were shared between all regions. Multidrug resistance genes were the most frequently encountered ARGs, followed by ARGs conferring resistance to β -lactams, aminoglycosides, tetracyclines, and macrolide-lincosamide-streptogramin B (MLSB). Research ponds managed by the Bangladesh Agricultural University had the lowest abundance and diversity of ARGs, suggesting that proper management such as regular water quality monitoring, fortnightly water exchange and use of probiotics instead of antibiotics may mitigate the dissemination of antibiotic resistance from aquaculture ponds. The Adonis test ($R^2 = 0.35$, $p < 0.001$) and distance decay relationships revealed that the ARGs composition displayed a significant biogeographical pattern (i.e., separation based on geographic origin). However, this effect could possibly be due to feed type as

* Corresponding author at: Section for Microbial Ecology and Biotechnology, Department of Plant and Environmental Sciences, University of Copenhagen, Thorvaldsensvej 40, 1871 Frederiksberg, Denmark.

E-mail address: kkb@plen.ku.dk (K.K. Brandt).

¹ Dual first authorship: Simon Bo Lassen and Md. Emranul Ahsan contributed equally to this study.

different feed types were used in different regions. In conclusion, our results indicate that there is a vast potential for improving aquaculture pond management practices in Bangladesh to mitigate the environmental dissemination of ARGs and their subsequent transmission to humans.

1. Introduction

Aquaculture has become a fast-growing industry due to the increasing demand for fish and seafood products throughout the world. It is estimated that around 156 million tons of fish and shellfish are consumed every year, with aquaculture production accounting for approximately 46% (FAO, 2020). To sustain the rapid and steady growth of this industry, antibiotics are widely applied, especially in low- and middle-income countries, to reduce infectious diseases and increase fish production (Holmström et al., 2003; Cabello et al., 2013; Zhao et al., 2020a). Consequently, aquaculture is considered a hot spot for environmental dissemination of antibiotic resistance and its transmission to humans via the food chain (Watts et al., 2017; Shen et al., 2018; Reverter et al., 2020). Fish pathogens and other aquatic bacteria have been reported resistant to multiple antibiotics due to the increased use of antibiotics (Watts et al., 2017; Sørsum et al., 2019; Schar et al., 2021). Furthermore, the presence of mobile genetic elements (MGEs) such as plasmids harboring antibiotic resistance genes (ARGs) allows for the horizontal gene transfer of ARGs between aquatic bacteria, fish pathogens, and even human-associated pathogens (Rhodes et al., 2000; Akinbowale et al., 2007; Sørsum, 2008; Marti et al., 2014). For instance, tetracycline resistance genes in bacteria isolated from three different fish farms showed high similarity with genes isolated from bacteria in humans, suggesting that ARGs in bacteria from fish farms can be transferred to human pathogens and potentially threaten human health (Furushita et al., 2003).

The majority of the global aquaculture production takes place in Asian countries, including Bangladesh, where aquaculture has significantly expanded over the last decade (Ali et al., 2016; Henriksson et al., 2018; Schar et al., 2020; Thorner et al., 2020). Bangladesh now ranks as the 5th highest aquaculture producing country in the world (FAO, 2020) with *Oreochromis niloticus* and *Pangasianodon hypophthalmus* as two of the major types of aquaculture products (Sumon et al., 2016; Shamsuzzaman et al., 2017; DoF, 2019). Moreover, Bangladesh currently ranks as one of the top consumers of antibiotics in aquaculture production (Lulijwa et al., 2019). Hence, an expanding range of antibiotics, including β -lactams, tetracyclines, and sulfonamides, are now widely used for disease treatment and growth promotion (Rico et al., 2012; Rico et al., 2013; Ali et al., 2016; Hinchliffe et al., 2018; Das et al., 2020). The antibiotics are administered by farmers, either by hand mixing the antibiotics with the feed or by direct application to the water surface during disease outbreaks (Ali et al., 2016). Most farmers related to aquaculture production, however, generally lack the proper knowledge on how to use antibiotics (Cabello, 2006; Ali et al., 2016). As a result, overuse or misuse of antibiotics in aquaculture often occurs (Sapkota et al., 2008; Heuer et al., 2009). In addition, several studies have shown that antibiotics administered by feed will enter the aquatic environment through excretions or leaching from uneaten medicated feed, and up to 80% of the antibiotics will remain in sediments or the water after application (Lalumera et al., 2004; Cabello et al., 2016).

Increasing global awareness and stricter regulations have led to a reduced application of antibiotics in many economically important production systems, especially those intended for the international market (Henriksson et al., 2015; Ali et al., 2016). To evaluate the effects of regulations and aquaculture practices on antibiotic resistance, national surveillance of ARGs should be conducted, which have already been implemented in some countries (Dejsirilert et al., 2009; Karp et al., 2017; Turnidge et al., 2019). In Bangladesh, studies have previously detected antibiotic-resistant bacteria and resistance determinants in fish ponds, including integrated aquaculture (Neela et al., 2015), shrimp aquaculture (Khan et al., 2007; Neela et al., 2012), and duckweed aquaculture (Rahman et al., 2009). However,

nationwide quantitative surveys of the antibiotic resistome in aquaculture systems in Bangladesh have not previously been carried out.

In this study, we initially collected water samples from 317 commercial *P. hypophthalmus* and *O. niloticus* aquaculture production ponds across eight districts representing four divisions (i.e. regions consisting of several districts) of Bangladesh. Based on an initial water quality assessment, we subsequently investigated the prevalence and diversity of ARGs in selected ponds ($n = 33$) across four districts representing four divisions of Bangladesh by a high-throughput quantitative PCR (HT-qPCR) chip system targeting 285 ARGs, 10 MGEs, and the bacterial 16S rRNA gene. Furthermore, we compared the prevalence of ARGs in commercial production ponds with best-case reference ponds with improved water quality control and use of probiotics instead of antibiotics. Specifically, we hypothesized that ARG and MGE abundance and diversity would be higher in commercial aquaculture ponds than in corresponding reference ponds.

2. Materials and methods

2.1. Sampling of aquaculture ponds from different districts in Bangladesh

Water samples were collected from 317 commercial *P. hypophthalmus* and *O. niloticus* aquaculture production ponds across eight districts of Bangladesh, namely Mymensingh, Bogura, Jashore, Khulna, Gopalganj, Madaripur, Chandpur, and Cumilla (Fig. S1). In addition, eight ponds constructed and used for research purposes by the Bangladesh Agricultural University (BAU) were sampled to constitute best-case scenario reference ponds (Islam et al., 2021). These reference ponds relied on probiotics rather than antibiotics for disease management and were strictly managed with regard to water quality with fortnightly water exchange. The reference ponds were reconstructed from pre-existing ponds located at the Faculty of Fisheries, BAU, Mymensingh, as described previously (Islam et al., 2021). Fish ponds were uniformly sized, each with a surface area of 40.5 m² and a depth of 1.20 m. To exclude undesirable fish species in the ponds they were sun dried and the bottom soil was replaced with soil sourced from non-agricultural land with no history of any farming or use of any kind of chemicals. The pond soil was conditioned with lime as is practiced in Bangladesh to improve the structure of the soil and to make the fertilizing materials act more efficiently in an alkaline condition. After two weeks, the ponds were filled with underground water up to 1 m depth. Separate inlet and outlet canals were built into each pond to prevent the exchange of fish between ponds, and dikes prevented the entry of other fish and the escape of stocks. No chemicals were applied and farm made formulated feed were used during the culture period (see Text S1 for details regarding feed composition, stocking density, probiotics used, etc.).

Water samples were collected from the surface of the aquaculture ponds at the end of the rainy season in September 2018 corresponding to the end of the aquaculture production cycle. Aquatic bacterial communities are in a constant state of flux, but by performing our sampling at the end of the fish production cycle we ensured that the sampled antibiotic resistomes would serve as the best possible indicators for assessing the impacts of previous aquaculture pond management. Three grab samples were taken from each pond using a water grab sampler (1 L capacity) (Ben Meadows Company, WI, USA) and were subsequently mixed and stored in pre-cleaned polypropylene (PP) bottles as one composite sample (a total of 1 L). The collected samples were transported in insulated plastic cooling boxes containing cooling elements. Upon arrival in the laboratory at the Bangladesh Agricultural University, the temperature of the samples was always below 15 °C. Samples were kept in the dark at 4 °C overnight and subsequently filtered through a 0.20 μ m pore size membrane filter to collect bacterial cells

(Djurhuus et al., 2017) using a filtration unit (GAST, Model: DOA-P504A-BN, Lab Model Diaphragm Pump, USA). The filters were subsequently dried and kept in the dark at 4 °C for subsequent DNA extraction. Samples to be used for bioreporter analysis (see below) were stored at −18 °C.

2.2. Aquaculture management information

Information on the management of the commercial aquaculture production systems was obtained via interviews with the farmers operating the ponds. Farmers were somewhat hesitant to fully participate in the interviews, and the information obtained was therefore somewhat limited, but it did include culture system type (mono or polyculture), feed type (commercial or farm-made), as well as types of antibiotics, fertilizers, and chemicals used in the ponds. Information regarding the eight reference ponds was provided by BAU.

2.3. Measurement of water quality parameters

A number of physicochemical water quality parameters were recorded for all samples in the field during sampling. Water temperature was recorded using a portable LCD Multi-Thermometer (Model: WT-2), and dissolved oxygen was measured by a digital DO meter (HANNA instruments, model: Lutron, DO-5509). The pH of the waters was measured by a digital pH meter (HANNA instruments, model: HI 98107). Other water quality parameters were measured in triplicates in the laboratory. Concentrations of nitrate, phosphate, and ammonia were measured in the laboratory by a direct reading spectrophotometer (Lovibond® PCspectro, model RS232, Dortmund, Germany, wavelength range from 330 to 900 nm) using Vario Nitrate Chromotropic powder pillows, Vario Phosphate RGT F10 powder pillow, and ammonia pillow, respectively.

Bioavailable tetracycline was determined in water samples using the bacterial biosensor strain *Escherichia coli* K12 (pTetLux1) (Korpela et al., 1998; Virolainen et al., 2008). In brief, overnight exponential phase biosensor cells were harvested and incubated with equal volumes of water sample for 3 h at 37 °C, 200 rpm in the dark. After incubation, bioluminescence was quantified using a BMG FluoStar Optima plate reader (BMG Labtech, Offenburg, Germany) as described previously (Song et al., 2017). The detection limit of the assay was operationally defined as a bioluminescence induction coefficient (IC) >2 (see Text S2 for details).

2.4. Selection of samples for HT-qPCR analysis of ARGs and MGEs

Only a small proportion (6.5%) of all samples contained tetracycline concentrations above the detection limit as determined by the whole-cell bioreporter assay (Table S1). We, therefore, randomly selected 25 samples, both with ($n = 6$) and without ($n = 19$) detectable tetracycline from four economically important aquaculture divisions with high production of *P. hypophthalmus* and *O. niloticus*, namely Mymensingh, Chattagram, Rajshahi, and Khulna (DoF, 2018). The divisions Mymensingh, Chattagram and Rajshahi have the highest production of *P. hypophthalmus* in Bangladesh, ranging between 54,431 up to 210,242 metric tons annually, while Chattagram, Khulna and Mymensingh have the highest production of *O. niloticus*, ranging between 43,647 up to 89,539 metric tons annually (DoF, 2018). The samples derived from four separate districts located within the four chosen divisions, namely Mymensingh ($n = 8$, Mymensingh Division), Cumilla ($n = 10$, Chattagram division), Bogura ($n = 3$, Rajshahi division), and Khulna ($n = 4$, Khulna division), and will be referred to by their district name throughout the paper. In addition, we included the eight samples from the BAU reference ponds managed with no use of antibiotics. Figure 1 depicts the geographical locations of the aquaculture ponds included in the present study. All sampled commercial aquaculture ponds had been constructed at land previously used for irrigated rice agriculture and relied on surface water with underground water being occasionally used only during the dry season. A power test was conducted to verify that the selected sample sizes provided enough statistical power to

compare means in relative abundance of ARGs between districts (see Text S3 for details).

2.5. Quantification of ARGs by HT-qPCR

High-quality DNA was extracted from the filters obtained in Section 2.1 using the DNeasy PowerWater® DNA Isolation Kit according to the instruction manual supplied by MO BIO Laboratories Inc., USA. HT-qPCR detection of ARGs was subsequently conducted using the Smart chip Real-time qPCR system (Wafergen biosystems) as described previously (Liu et al., 2018). A total of 296 primer pairs were used, targeting 285 ARGs, 10 MGEs, and the 16S rRNA gene (Ding et al., 2020). The amplification efficiency was 90–110%, and the CT threshold for detection of target genes was 31. All qPCR analyses were conducted with analytical triplicates, and results were only included if target genes could be detected in all analytical triplicates. The relative abundance of each ARGs and MGEs was used in this study, and was calculated according to the following equation (Looft et al., 2012): ARG Copy Number = $10^{(31-C_T)/(10/3)}$.

The ARG copy numbers were subsequently normalized to the 16S rRNA gene copy number.

The absolute abundance (genes/L) of each ARGs in the ponds was calculated by first quantifying the absolute abundance of the 16S rRNA gene in each sample using SYBR® Green on the Lightcycler 480 instrument (Roche, Basel, Switzerland) as described by (Ouyang et al., 2015). As there was a high correlation between the relative (SmartChip) and absolute (Lightcycler) abundance of the 16S rRNA gene ($r = 0.83$, $p < 0.001$), we transformed the relative abundance of ARGs to absolute abundance according to previous studies (Ouyang et al., 2015; Liu et al., 2018).

2.6. Network analysis and visualization

The software Cytoscape using the CoNet plugin was used to explore the co-occurrence pattern among ARGs and MGEs, as described previously (Soffer et al., 2015; Hu et al., 2017). In brief, interaction scores were calculated using Spearman and Pearson correlation methods and Kullback-Leibler and Bray-Curtis dissimilarity methods. A reboot procedure with 100 permutations was conducted to eliminate false-positive correlations, and all obtained p -values were merged using the Brown method, and only connections that were significant in at least two of the methods were included in the network. A significant pairwise correlation coefficient above 0.6 and a merged p -value below 0.05 was used for constructing the co-occurrence network.

2.7. Statistical analysis

Significant differences in the abundance of ARG and MGE classes (grouped according to class of antibiotic they confer resistance to or class of MGE) between the commercial ponds and the reference ponds were assessed using the Kruskal–Wallis rank test combined with Dunn's test for multiple comparisons (Kruskal and Wallis, 1952; Dunn, 1964). The Benjamin-Hochberg method was used for recalculating and adjusting the obtained p -values (Benjamini and Hochberg, 1997). Non-metric multidimensional scaling (NMDS) was performed based on relative abundances of ARGs to investigate differences in the overall pattern of ARGs between the districts. The Adonis test (Anderson, 2001) was performed to identify significant differences in ARG patterns based on Bray-Curtis distance matrices. The respective district of the ponds and type of feed were used as distinguishing variables for the Adonis test. Redundancy analysis (RDA) was used to investigate the relationship between the relative composition of ARGs, environmental factors (ammonia, nitrate, phosphate, dissolved oxygen, pH, and bioavailable tetracycline), pond characteristics (depth, age, and feed-type), and relative abundance of MGEs of the ponds. Significant correlations between variables were determined by Spearman's rank correlation test. All statistical analyses were conducted using R version 4.0.3 (2020-10-10). The Adonis test and RDA were conducted using the vegan package. Distance decay analysis of dissimilarities in relative ARG

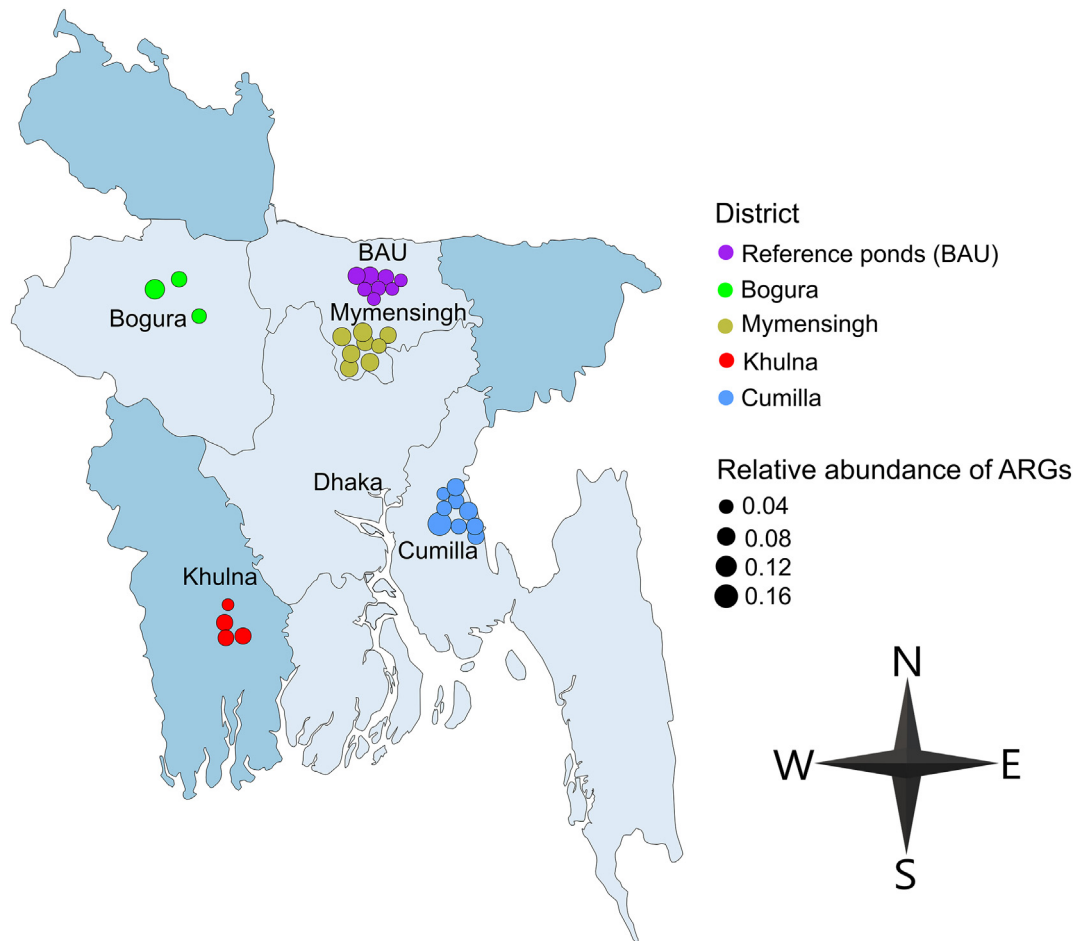


Fig. 1. Map of Bangladesh showing the sampling locations of aquaculture ponds used for investigating the prevalence of antibiotic resistance genes (ARGs) and mobile genetic elements (MGEs). The relative abundance of ARGs (ARGs per 16S rRNA gene) is indicated by the size of the closed circles. All sampled ponds are named and colored according to the districts in which they are located, with the addition of the reference ponds maintained by Bangladesh Agricultural University (BAU). The relative abundances of ARGs and MGEs were quantified in ponds from BAU ($n = 8$), Bogura ($n = 3$), Mymensingh ($n = 8$), Khulna ($n = 4$), and Cumilla ($n = 10$). Cardinal direction is indicated by the compass located in the bottom right of the figure.

compositions and geographical distances between ponds were conducted using the betapart package. The map of Bangladesh was constructed using the R package choroplethr in combination with Paint 3D.

3. Results

3.1. Characterization of aquaculture ponds used for HT-qPCR analysis

Aquaculture pond management practices were retrieved by interviewing aquaculture farmers, and an overview of the interview findings is presented in Table 1. Most of the farmers continued the production cycle throughout the year, except for the winter season. Surface water was the main source of water for most of the ponds; however, some ponds also received underground water during the dry season. The water exchange rate was very low in the surveyed commercial ponds, and in most cases, there was no water exchange during the entire production cycle. Most ponds from the northern districts of Bangladesh were given farm-made feed, while the ponds of the southern districts were given commercial feed. Ingredients in farm-made feed typically include rice bran/rice polish (30–60% dry wt), maize, fish meal, meat and bone meal, mustard oil cake, soybean meal, salt, and vitamins, whereas commercial feed typically has a high protein content around 30% and is prepared from a mixture of animal products (e.g. fish meal and bone meal), plant products (e.g. maize, soybean meal, rice bran, mustard oil cake and wheat flour), vitamins, and minerals (NaCl and various trace elements) (Ali et al., 2013).

According to the interviewed aquaculture farmers, different types of antibiotics, including ciprofloxacin, amoxicillin trihydrate, tetracycline, oxy-tetracycline, and chlortetracycline, were used at different stages of the fish production to prevent infectious diseases and promote growth. Besides, most farmers applied lime and salts (see Table 1) to the ponds between production cycles, when the ponds were drained for water, and some farmers also applied lime and salts directly to the water to kill potential pathogens and pests, especially before winter to control disease or parasite outbreak. Urea and Triple superphosphate were the most used fertilizers in the surveyed ponds and were used to boost phytoplankton production as a source of fish feed. Farmers also applied cow manure to some of the ponds, which can be directly consumed by the fish and release nutrients supporting phytoplankton growth.

In contrast, the reference ponds were subjected to a 50% water exchange every fortnight, and did not receive any antibiotics. Instead, ponds were amended with probiotics such as *Bacillus subtilis* and *Rhodopseudomonas* sp. to combat diseases and promote growth (Text S1). In addition, all reference ponds used in this study were fed farm-made feed, which was composed of fish meal (25%), soybean meal (25%), rice bran (24%), wheat flour (24%), salt (1%), and vitamin-mineral premix (1%) (Text S1). The farm-made feed of BAU was thus different from the typical farm-made feed used by commercial aquaculture operations (Ali et al., 2013).

The concentration of bioavailable tetracyclines in all surveyed ponds was mostly below the detection limit of the biosensor assay, but 21 samples (6.5%), distributed across different districts, contained high concentrations

Table 1

General management information as obtained via interviews with respective aquaculture farmers for the aquaculture ponds selected for HT-qPCR analysis.

District	Culture system ^a	Feed type	Fertilizer ^b	Water source	Water exchange	Water & soil treatment compounds ^c	Disinfectants ^d	Antibiotic/probiotic	O ₂ supplement ^e
Reference ponds (BAU)	Pangasius MC/tilapia MC	Farm made	-	Underground/surface water	Fortnightly 50% water exchange	Lime ^f /zeolite	-	<i>Bacillus subtilis</i> / <i>Rhodospseudomonas</i>	2N ₂ CO ₃ × 3H ₂ O ₂
Bogura	Pangasius IPC/tilapia SMC	Commercial/farm made	Urea/TSP/cow manure	Underground/surface water	Low or no water exchange	Lime ^f /NaCl/K ₂ Cr ₂ O ₇ /zeolite/CuSO ₄	H ₂ O ₂ /ADBAC/BKC	OTC/TC/ciprofloxacin	2N ₂ CO ₃ × 3H ₂ O ₂
Mymensingh	Pangasius IMC/tilapia SMC	Farm made	Urea/TSP/cow manure	Underground/surface water	Low or no water exchange	Lime ^f /NaCl/K ₂ Cr ₂ O ₇ /zeolite/CuSO ₄	H ₂ O ₂ /BKC/PP	OTC/TC/DC	2N ₂ CO ₃ × 3H ₂ O ₂
Khulna	Pangasius IPC/tilapia SMC	Commercial	Cow manure	Underground/surface water	Low or no water exchange	Lime ^f /NaCl/K ₂ Cr ₂ O ₇ /zeolite/CuSO ₄	H ₂ O ₂ /TTAB/PP	Amoxicillin trihydrate/OTC	2N ₂ CO ₃ × 3H ₂ O ₂
Cumilla	Pangasius IPC/tilapia SMC	Commercial	Urea/TSP/cow manure	Underground/surface water	Low or no water exchange	Lime ^f /NaCl/K ₂ Cr ₂ O ₇ /zeolite/CuSO ₄	H ₂ O ₂ /PP/ADBAC	CTC/ciprofloxacin	2N ₂ CO ₃ × 3H ₂ O ₂

Abbreviations: BAU = Bangladesh Agricultural University, MC = monoculture, IPC = intensive polyculture, SMC = semi-intensive monoculture, IMC = intensive monoculture, TSP = Triple superphosphate, ADBAC = n-Alkyl dimethyl benzyl ammonium chloride (Chlorine releasing compound), BKC = Benzalkonium chloride, PP = Potassium permanganate, TTAB = Tetradecyl trimethyl ammonium bromide, OTC = Oxytetracycline, TC = Tetracycline, DC = Doxycycline, CTC = Chlorotetracycline.

^a See Jahan et al., 2015 for a detailed definition of the culture systems.

^b Fertilizers are used during pond preparation and maintenance to boost growth of phytoplankton to be consumed by fish.

^c All farmers used at least one or two substances for water or soil treatment in between the production cycles or directly into the water during the production period.

^d Disinfectants are used to disinfect culture facilities and equipment, sediment, and pond water to control bacterial disease outbreak.

^e Oxygen releasing compound used to increase the dissolved oxygen level in pond water.

^f The most commonly used water and soil treatments are liming compounds (calcium oxide and calcium carbonate).

of tetracyclines (Table S1). No bioavailable tetracyclines were detected in the reference ponds.

3.2. Diversity and abundance of ARGs and MGEs in aquaculture ponds

A total of 160 unique ARGs and 10 MGEs were detected across all studied aquaculture ponds of this study (Fig. S2). Compared to the reference ponds maintained by BAU, a significantly higher number of ARGs was detected in the ponds from the Mymensingh ($p = 0.043$) and Khulna ($p = 0.045$) districts, while no significant differences were found for the number of MGEs (Fig. 2A). Among all ARGs and MGEs detected, 76 were shared between all districts, including BAU (i.e., present in at least one pond per district), while 34 were shared between all ponds.

The detected genes could confer resistance to most major classes of antibiotics (Fig. 2B), with the highest percentage of genes conferring resistance to multidrug (27%) followed by β -lactams (17%), aminoglycosides (15%), tetracycline (12%) and macrolide-lincosamide-streptogramin B (MLSB) (10%). The mechanism of the ARGs mainly constituted efflux pumps (40.6%), antibiotic deactivates (38.1%), and cellular protection (12.6%) (Fig. S3).

The accumulative relative abundance of ARGs (i.e., sum of all ARGs abundances) was lower in the reference ponds maintained by BAU compared to the commercial aquaculture ponds, (Figure 3A), but the difference was not statistically significant. However, a significantly higher abundance of β -lactam resistance genes ($p = 0.01$) was detected in the Cumilla district (0.017 ± 0.008) compared to the reference ponds (0.005 ± 0.003). Furthermore, the Mymensingh district had a significantly higher relative abundance of multidrug resistance genes (0.04 ± 0.008 ; $p = 0.02$) and MLSB genes (0.004 ± 0.001 ; $p = 0.03$) compared to the reference ponds (0.02 ± 0.009 and 0.002 ± 0.001 , respectively).

The absolute abundance of ARGs was significantly lower in the reference ponds than those of all commercial districts ($p < 0.05$), which was also the case for multidrug resistance genes ($p < 0.05$) (Fig. 3C). Moreover, the absolute abundance of integron genes were significantly ($p < 0.05$) lower in the reference ponds ($3 \times 10^8 \pm 8 \times 10^6$) compared to the Bogura ($1 \times 10^9 \pm 1 \times 10^8$) and Mymensingh ($1 \times 10^9 \pm 2 \times 10^8$) districts (Fig. 3D). The absolute abundance of sulfonamide resistance genes was, in addition, significantly higher ($p < 0.05$) in the Bogura ($6 \times 10^8 \pm 2 \times 10^8$) and Mymensingh ($4 \times 10^8 \pm 3 \times 10^8$) districts compared to the reference ponds ($4 \times 10^7 \pm 3 \times 10^7$). Moreover, a significantly ($p < 0.05$) higher abundance of β -lactam resistance genes was detected in the ponds of the Cumilla ($1 \times 10^9 \pm 5 \times 10^8$) and Khulna ($8 \times 10^8 \pm 5 \times 10^8$) districts, compared to the reference ponds ($2 \times 10^8 \pm 7 \times 10^7$). In addition, the absolute abundance of vancomycin resistance genes was significantly ($p < 0.05$) lower in the reference ponds ($5 \times 10^7 \pm 2 \times 10^7$) compared to the Cumilla ($1 \times 10^8 \pm 3 \times 10^7$), Khulna ($1 \times 10^8 \pm 5 \times 10^6$), and Mymensingh ($1 \times 10^8 \pm 6 \times 10^7$) districts.

3.3. Correlations between ARGs, MGEs, and environmental factors

Significant correlations between the accumulative abundances of ARGs and MGEs were detected (Figs. S4 and S5; Tables S2 and S3). The clinical integrase gene *Int1* showed a significant correlation with ARGs against MLSB, multidrug, pyrazinamide, sulfonamide, and vancomycin at both the absolute and relative abundance levels, and correlated with ARGs against FCA (fluoroquinolone, quinolone, florfenicol, chloramphenicol, and amphenicol) at the absolute abundance level (Tables S4 and S5). The transposase gene *Tnp-4* correlated significantly with ARGs against FCA, aminoglycoside, β -lactams, sulfonamides, and tetracycline at both relative and absolute abundance levels. Furthermore, our network analysis revealed co-occurrence relationships between several ARGs and MGEs (Fig. S6). For instance, a cluster consisting of the multidrug resistance gene *qacEdelta1*, the florfenicol resistance gene *florR*, the tetracycline resistance *tetG*, the sulfonamide resistance gene *sul2*, and several aminoglycoside resistance genes, showed strong connections to the MGEs, *tnpA-04* and *Int1*, suggesting a potential for mobilization of these ARGs.

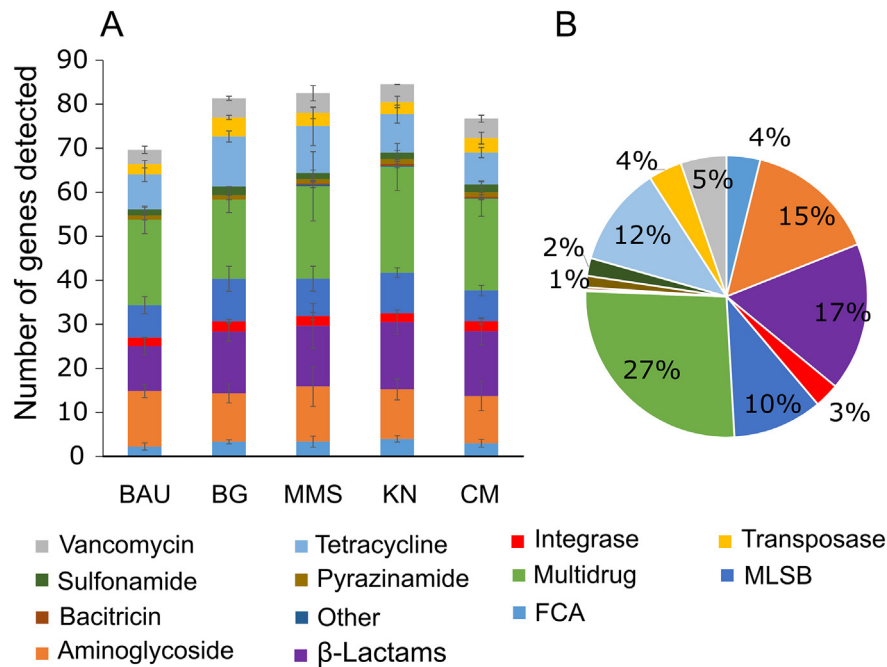


Fig. 2. (A) Number of detected antibiotic resistance genes (ARGs) and mobile genetic elements (MGEs) in aquaculture ponds from different districts in Bangladesh: BAU – reference ponds maintained by Bangladesh Agricultural University $n = 8$, BG - Bogura $n = 3$, MMS - Mymensingh $n = 8$, KN - Khulna $n = 4$, and CM - Cumilla $n = 10$. Data are presented as mean $\pm$ standard deviation. (B) The overall distribution of ARGs and MGEs in all ponds categorized by the class of antibiotics. FCA: fluoroquinolone, quinolone, florfenicol, chloramphenicol, and amphenicol resistance genes; MLSB: Macrolide-Lincosamide-Streptogramin B resistance genes.

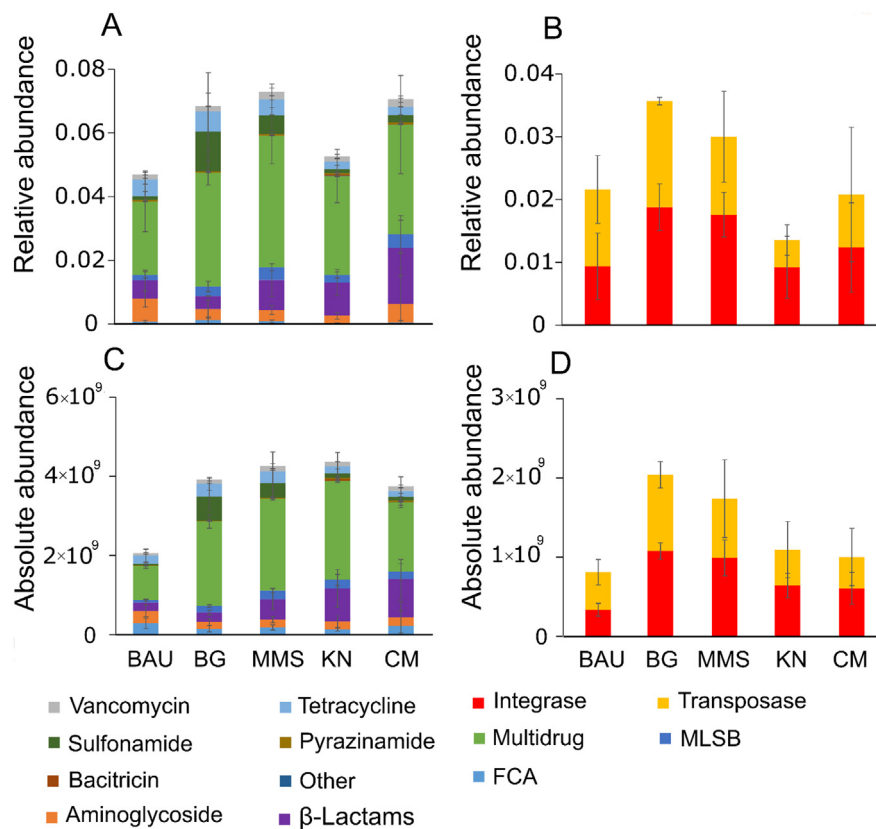


Fig. 3. Relative abundance (Panels A and B) and absolute abundance (Panels C and D) of antibiotic resistance genes (ARGs; Panels A and C) and mobile genetic elements (MGEs; Panels B and D) in aquaculture ponds from the different districts in Bangladesh: BAU - reference ponds maintained by Bangladesh Agricultural University $n = 8$, BG - Bogura $n = 3$, MMS - Mymensingh $n = 8$, KN - Khulna $n = 4$, and CM - Cumilla $n = 10$. FCA: fluoroquinolone, quinolone, florfenicol, chloramphenicol, and amphenicol resistance genes; MLSB: Macrolide-Lincosamide-Streptogramin B resistance genes. Data are presented as mean $\pm$ standard deviation.

Several environmental parameters and characteristics of the ponds (Table S6) correlated with the abundance of ARGs. The relative and absolute abundance of ARGs significantly correlated with both the age and depth of the ponds, while the absolute abundance of ARGs also correlated with water temperature, DO, and phosphate content (Table 2). Several correlations were also detected between specific ARGs classes and the quantified pond parameters and characteristics (Tables S7 and S8).

3.4. Relationship between environmental variables and ARG/MGE distribution patterns

The composition of ARGs in each pond could be separated by their respective district ($R^2 = 0.35, p < 0.001$) and more weakly by cardinal direction (North vs. South) ($R^2 = 0.18, p = 0.001$), or feed type used (farm-made vs. commercial) ($R^2 = 0.17, p = 0.001$) as determined by the Adonis test (Fig. 4). Furthermore, the ARGs showed a significant distance decay relationship, indicating that ponds located close together had a more similar ARG composition compared to ponds separated by larger distances (Fig. S7).

Redundancy analysis (RDA) was conducted to investigate the relationship between environmental variables, feed-type, and ARG compositions (Fig. S8). The results revealed that 61.7% of the variance in the relative ARG compositions could be explained by the measured variables, with MGEs, pH, and feed type being the most significant factors associated with the ARG compositions.

4. Discussion

4.1. ARGs in aquaculture ponds from different districts of Bangladesh

To our knowledge, this is the first comprehensive investigation of ARGs in aquaculture systems across multiple districts of Bangladesh. Our results revealed a high diversity and prevalence of ARGs and MGEs across all investigated districts of Bangladesh. The reference ponds maintained by BAU had the lowest diversity and abundance of ARGs, which is likely a result of the proper management, including feed preparation under hygienic conditions, regular exchange of water, use of probiotics instead of antibiotics, and more prudent use of chemicals. Hence, our results indicate that there is a potential for improving aquaculture pond management practices in Bangladesh to mitigate the environmental dissemination of ARGs and their subsequent transmission to humans e.g., via the food chain.

Several other studies have documented antibiotic resistance in aquaculture in different parts of the world. A recent review focused on antibiotic use and associated environmental and public health risks in the top 15 aquaculture-producing countries in the world (Lulijwa et al., 2019). Out of the 170 case studies included in the review, most studies were conducted in China (74), followed by Vietnam (21) and India (20), while only two case studies were conducted in Bangladesh. One of the Bangladesh studies used antibiotic concentrations to estimate risk quotients (RQs) in aquaculture ponds across Bangladesh, but did not find any indication that the concentrations present constituted a risk for antibiotic resistance development

Table 2

Spearman's rank correlations between environmental variables and the relative (Rel.; copies/16s rRNA gene) and absolute (Abs.; copies/L) abundance of antibiotic resistance genes (ARGs).

Environmental variables	Range	Rel. ARGs	p-Value	Abs. ARGs	p-Value
Age of pond (years)	5.0–18.0	0.41	0.017	0.51	0.002
Avg. depth (m)	1.0–2.45	0.46	0.005	0.61	0.000
Water temp. (°C)	28–31	0.23	0.2	0.74	8E–07
pH	6–9.4	0.26	0.15	0.63	0.00
Dissolved O ₂ (mg L ⁻¹)	4.1–7.6	0.02	0.91	0.50	0.003
Nitrate (mg L ⁻¹)	0–9.3	0.02	0.90	–0.17	0.35
Phosphate (mg L ⁻¹)	0.0–1.1	0.14	0.43	0.41	0.016
Ammonia (mg L ⁻¹)	0.0–3.7	0.12	0.50	0.17	0.34

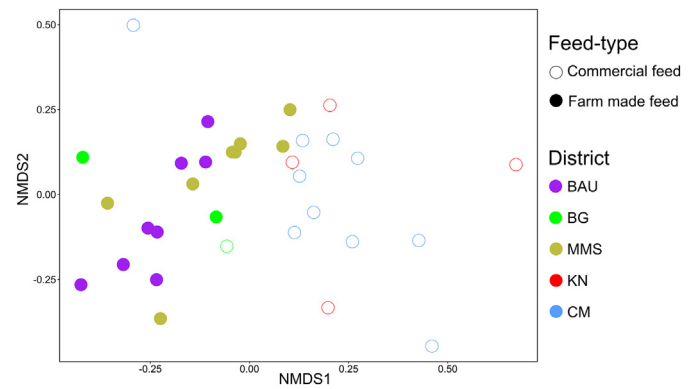


Fig. 4. Non-metric multidimensional scaling (NMDS) ordination of the relative antibiotic resistance gene (ARG) composition in aquaculture ponds from different districts in Bangladesh. Samples are colored by district: BAU - reference ponds maintained by Bangladesh Agricultural University $n = 8$, BG - Bogura $n = 3$, MMS - Mymensingh $n = 8$, KN - Khulna $n = 4$, and CM - Cumilla $n = 10$. Feed-type administered to the ponds are indicated by open (Commercial feed) or closed (farm-made) circle.

(Hossain et al., 2017). Another study from a duckweed aquaculture system heavily affected by hospital sewage water in Bangladesh detected high levels of antibiotic resistance in *Aeromonas* strains isolated from aquaculture ponds with bacteria exhibiting similar resistance levels as comparable strains isolated from wastewater (Rahman et al., 2009). Our results indicate that antibiotics might have had a direct effect on antibiotic resistance, as antibiotics only were used in the commercial ponds of this study, which harbored the highest abundance and diversity of ARGs. However, other factors such as water-exchange rate and inputs of fecal matter (e.g., cow manure used as fertilizers or fecal contamination) are also likely to contribute (Karkman et al., 2019; Thongsamer et al., 2021). Moreover, water from 6.5% of all analyzed ponds ($n = 325$) of this study induced tetracycline stress in *E. coli* as witnessed from activation of *luxCDABE* gene expression under control of the *tetA* promoter. However, our biosensor data just represents a 'snapshot' of tetracycline exposure as concentrations of tetracyclines will change over time depending on local aquaculture farm management conditions. Quantification of antibiotic concentrations alone may therefore not be the best approach to assess antibiotic resistance in aquaculture, unless high-temporal resolution studies can be conducted. Hence, we focused on detection of ARGs and MGEs as these antibiotic resistance determinants can be considered integrated effect parameters for aquaculture management and water quality.

A total of two integrons and eight transposon genes were targeted and all were detected in this study. Both integrons (*int1-1-clinic* and *int1-1LC*) and the transposase *tnpA-04* were detected in all ponds. Several significant correlations were observed between the individual MGE genes and classes of ARGs (Tables S4 and S5). The strong correlations suggest that at least some ARGs were located on MGEs, thereby increasing the risk for horizontal gene transfer of ARGs. Through redundancy analysis, we furthermore discovered that MGEs were strongly associated with the composition of ARGs, further indicating a tight link between MGEs and ARGs. Similar results have been observed in estuarine sediments in China, in which microbial community composition and MGEs were the major drivers for variations in the ARG composition (Zhao et al., 2020b). Plasmids carrying ARGs in aquatic fish pathogens are not uncommon and have been frequently detected (Dixon et al., 1990; Preena et al., 2020; Shen et al., 2020). For instance, incompatible IncA/C host range plasmids containing the *qacE2*, *tetA*, *tetD*, *tetE* and *floR* genes along with various aminoglycoside resistance genes have previously been isolated from ornamental fishes (Verner-Jeffreys et al., 2009). In our co-occurrence network analysis (Fig. S6), we observed strong correlations between the *qacE2* homolog *qacEdelta1*, *floR*, *tetG*, several aminoglycoside resistance genes, and the transposase gene *tnpA-04*, which could indicate a similar co-occurrence of these genes as observed by Verner-Jeffreys et al. Another study investigating

plasmids in antibiotic-resistant isolates from Chilean salmon, observed that the *qacEdelta1* and *sul1* genes frequently co-occurred in strains carrying *Int11* integrons, and observed a high transferability of the *sul* genes in mating experiments (Domínguez et al., 2019). In our study, we observed strong correlations between the *qacEdelta1*, *sul2*, and *Int11* genes (Fig. S6), which may indicate a similar co-location and transferability, as observed in the Chilean salmon study. The clinical *int11* integron, furthermore, exhibited a strong correlation with several ARGs, which could suggest a potential for mobilization and dispersal of these genes. The occurrence of ARGs on MGEs poses a risk for humans consuming the fish, as the ARGs might be passed on in the food chain (Shen et al., 2020; Zhao et al., 2020a).

The composition of ARGs in the ponds exhibited a biogeographical pattern, which could be separated based on the respective pond districts (Fig. 4). Interestingly we noted that the ARG compositions also could be weakly separated by Northern (Mymensingh, Bogura and BAU) and Southern (Cumilla and Khulna) districts ($R^2 = 0.16$, $p = 0.001$). We speculate that the observed separation between Northern and Southern ponds likely is due to differences in pond management as most ponds from the North were fed farm-made feed, while southern ponds were given commercial

feed (Farm-made vs. commercial, $R^2 = 0.15$, $p = 0.001$). Through redundancy analysis, we furthermore found that the feed-type/location of the ponds, along with MGEs and pH, were the most significant factors associated with ARG compositions (Fig. S8).

4.2. Comparison with other studies using the Wafergen SmartChip real-time PCR system in aquatic environments

In order to tentatively rank the risk associated with ARGs in aquaculture systems of Bangladesh, we reviewed the literature for all aquatic studies utilizing the same HT-qPCR chip as used in this study (Table 3). Only three of these studies focused on aquaculture facilities, including a study of aquaculture sediments in Finland, a fish farming area in Ying lake of China and coastal aquaculture systems in Central Thailand (Muziasari et al., 2016; Gu et al., 2019; Thongsamer et al., 2021). In addition, a fourth microcosm study focused on mariculture sediment spiked with tetracycline and/or fishmeal (Han et al., 2017). The number of ARGs detected in the aquaculture systems in Finland and Ying lake (ranging between 46–55 and 36–59 ARGs, respectively) was somewhat lower than both the commercial Bangladeshi ponds (ranging between 55 and 98 ARGs), and the reference

Table 3

Reduced table of studies investigating antibiotic resistance genes (ARGs) in aquatic environments using the Wafergen Smartchip Real-Time PCR system. See Table S9 for the full table.

Sample type	Source	No. of ARGs detected/no. of ARGs targeted	Relative abundance of ARGs (genes/16s rRNA gene)	Absolute abundance of ARGs (genes/L)	Methods (DNA extraction kit, qPCR cycles, and CT cutoff value)	Reference
East Xiaoxi river, China	Upstream	84/285	0.008	$<1 \times 10^9$	FastDNA SPIN Kit, 40 qPCR cycles, CT = NA	(Zheng et al., 2017)
	Agriculture or industrial impacted areas	105–142/285	0.013–0.04 ^c	1×10^9 – 2.1×10^{10}		
Urban stream of Jiulongjiang River, China	Upstream (pristine)	79/285	0.065 ^a	$\sim 10^9$	FastDNA SPIN Kit, 40 qPCR cycles, CT = 31	(Ouyang et al., 2015)
	Downstream (urban)	131–161/285	$\sim 0.6^a$	$\sim 10^{11}$		
Peri-urban Zhangxi river, Ningbo, China	River	46–154/285	~ 0.0025 (upstream) - 0.09 (downstream) ^c	1.35×10^8 (upstream) - 3.93×10^{10} (downstream)	FastDNA SPIN Kit (MP Bio, USA), 40 qPCR cycles, CT = 31.	(Zheng et al., 2018)
Urban river, Wenzhou, China	Main river and branches	100–159/285	0.017–0.2 ^c	3.61×10^{10} – 3.02×10^{11}	FastDNA SPIN Kit, 40 qPCR cycles, CT = 31	(Zhou et al., 2017)
Water bodies in China	42 Chinese Lakes and reservoirs	15–57/285	Avg. 0.015 (s.e. 0.002)	5.95×10^6 – 1.85×10^9	FastDNA SPIN Kit, 40 qPCR cycles, CT = 31	(Liu et al., 2018)
Estuaries of eastern China	18 estuaries across seven eastern Chinese provinces	85–158/285	~ 0.05 – 0.5^a	$\sim 8.8 \times 10^5$ – 1.8×10^8 ^e	FastDNA SPIN Kit, 40 qPCR cycles, CT = 31	(Zhu et al., 2017)
Estuarine sediments, Bohai Bay, China	Upstream tidal gate (eDNA)	$\sim 20/85$	0.0127	9.06×10^6 ^e	eDNA extracted using 1% cetyltrimethyl-ammonium bromide (CTAB). iDNA extracted by Soil DNA Extraction Kit (Omega, USA). 40 qPCR cycles, CT = 31.	(Zhao et al., 2020b)
	Upstream tidal gate (iDNA)	$\sim 32/85$	0.3	2.93×10^8 ^e		
	Estuary (eDNA)	~ 13 – $33/85$	~ 0.03 – 0.136	$\sim 2.5 \times 10^7$ – 1.32×10^8 ^e		
	Estuary (iDNA)	~ 25 – $48/85$	~ 0.02 – 0.03	$\sim 2.5 \times 10^7$ – 6.0×10^7 ^e		
Fish farming in Ying lake, China	Fenced fish farming area	36/284	~ 0.08	NA	E.Z.N.A. Water DNA kit, 40 qPCR cycles, CT = 31.	(Gu et al., 2019)
	box-type fish farming area	59/284	~ 0.5	NA		
	Control area	45/284	~ 0.0008	NA		
Aquaculture sediment in Finland	Aquaculture sediment	46–55/285	0.047–0.057	NA	FastDNA SPIN Kit, 40 qPCR cycles, CT = 27	(Muziasari et al., 2016)
	Surrounding sediment in Baltic sea	45/285	0.064–0.21	NA		
Mariculture sediment in microcosms treated with fishmeal and/or tetracycline (tet)	Untreated	$\sim 40/284$	$\sim 0.012^b$	$\sim 1.5 \times 10^6$ ^e	FastDNA SPIN Kit, 40 qPCR cycles, CT = 31	(Han et al., 2017)
	Fishmeal amended	$\sim 50/284$	$\sim 0.02^b$	$\sim 4 \times 10^6$ ^e		
	Tet amended	~ 45 – $50/284$	~ 0.01 – 0.012^b	~ 1.4 – 2×10^6 ^e		
	Fishmeal + tet amended	~ 50 – $60/284$	~ 0.017 – 0.03^b	~ 3 – 6×10^6 ^e		
Coastal aquaculture in Central Thailand	Aquaculture water	NA/283	~ 0.03 – 0.13	$\sim 1.9 \times 10^9$ – 1.1×10^{10} ^d	PowerWater® DNA Isolation Kit, 40 qPCR cycles, CT = 31	(Thongsamer et al., 2021)
	Aquaculture sediment	NA/283	~ 0.05 – 0.09	$\sim 1.1 \times 10^8$ – 4.3×10^8 ^e		
	Adjacent canal water	NA/283	~ 0.03 – 0.27	$\sim 3.9 \times 10^8$ – 1.8×10^{10} ^d		
	Adjacent canal sediment	NA/283	~ 0.04 – 0.11	$\sim 4.8 \times 10^7$ – 1.3×10^9 ^e		

^a Relative abundance is recalculated from genes/bacterial cell to genes/16s rRNA gene using the conversion number of 4.0 copies of 16s rRNA genes per bacterial cell, as stated in the study.

^b Relative abundance is recalculated from genes/bacterial cell to genes/16s rRNA gene using the conversion number of 4.1 copies of 16s rRNA genes per bacterial cell, as stated in the study.

^c Relative abundance is recalculated from genes/bacterial cell to genes/16s rRNA gene using the conversion number of 4.0 copies of 16s rRNA genes per bacterial cell. The conversion number used in the study was not clearly stated.

^d Absolute abundance of ARGs is recalculated from genes/mL, as indicated in the study to genes/L.

^e Absolute abundance of ARGs is determined as genes per gram of sediment.

ponds (50–78 ARGs). The number of ARGs detected in the coastal aquaculture systems of Central Thailand was not reported by the study. The accumulative relative abundance of ARGs in the commercial Bangladeshi ponds (ranging between 0.03 and 0.16 ARGs per 16S rRNA gene) was similar to fence-type fish farming in Ying lake (~0.08 ARGs per 16S rRNA gene) and the coastal aquaculture in Thailand (0.03–0.13 ARGs per 16S rRNA gene), but lower than box type fish farming in Ying Lake (~0.5 ARGs per 16S rRNA gene). On the contrary, the relative abundance of ARGs in the reference ponds (0.03–0.07 ARGs per 16S rRNA gene) was more similar to the sediment of Finnish aquaculture (0.047–0.057 ARGs per 16S rRNA gene), where relatively low amounts of antibiotics were presumably used due to European regulations (Preena et al., 2020). However, in the Finnish study, a higher relative abundance of ARGs was detected in the surrounding background sediment (0.064–0.21 ARGs per 16S rRNA gene), whereas aquaculture production had a more significant impact on the ARG composition. In addition, the study from Central Thailand reported a higher abundance of ARGs in canals adjacent to the aquaculture systems (~0.03–0.27 ARGs per 16S rRNA gene) than the aquaculture systems themselves due to fecal contamination from surrounding cities. The study thus concludes that there was no indication that aquaculture constitutes a major driver of environmental antibiotic resistance spread in Central Thailand's coastal district. Compared to other aquatic environments, the relative abundance of ARGs in this study was at a similar range as several anthropogenic-impacted rivers (ranging between 0.013 and 0.2 ARGs per 16S rRNA gene) (Zheng et al., 2017; Zheng et al., 2018; Zhou et al., 2017) and estuaries of China (~0.05–0.5 ARGs per 16S rRNA gene) (Zhu et al., 2017), but higher than suburban water bodies of China (Avg. 0.015 ARGs per 16S rRNA gene) (Liu et al., 2018).

The absolute abundance of ARGs in the commercial Bangladeshi ponds (ranging between 2.1×10^9 – 7.8×10^9 genes/L), was similar to the absolute abundance of ARGs detected in the aquaculture systems and adjacent canals of Central Thailand (~ 1.9×10^9 – 1.1×10^{10} and 3.9×10^8 – 1.8×10^{10} , respectively). In contrast, the absolute abundance of ARGs was higher in the Bangladeshi ponds than suburban water bodies (5.95×10^6 – 1.85×10^9 genes/L) and estuaries (~ 8.8×10^5 – 1.8×10^8 genes/L) of China, while somewhat similar abundances were detected in the reference ponds (1.3×10^9 – 2.8×10^9 genes/L). Compared to the absolute abundances reported for the anthropogenic-impacted Chinese rivers (1.0×10^9 – 3.02×10^{11} genes/L), the Bangladeshi ponds were mainly in the lower range, suggesting a higher microbial abundance in the Chinese rivers. Not surprisingly, there was a higher abundance of ARGs in wastewater treatment systems than the aquaculture ponds of this study (Table S9). Especially in influent samples, which is likely due to the high input of fecal waste products containing high levels of selective agents, resistance genes and microbes (Pärnänen et al., 2019). Even though we have attempted to compare findings across multiple HT-qPCR studies, it should be mentioned that this comes with some uncertainties due to a lack of method standardization. Nevertheless, our findings suggest that antibiotic resistance in commercial aquaculture production systems in Bangladesh constitutes a challenge that must be dealt with in the future.

5. Conclusions

Overall, we observed high diversity of ARGs in commercial *P. hypophthalmus* and *O. niloticus* aquaculture production systems across Bangladesh. Our results indicate that proper management such as regular water exchange and use of probiotics rather than antibiotics can significantly mitigate proliferation and dissemination of ARGs as observed for the reference ponds. The practices of BAU could thereby serve as a useful guideline for Bangladeshi aquaculture production in the future, although more studies are required to fully evaluate the involved management procedures. The ARG compositions of the ponds displayed a biogeographical pattern, and our results suggest that this pattern may be related to different feed-type used in different districts. Compared to other studies using the HT-qPCR smart chip ARG system, the abundance of ARGs in the aquaculture

ponds was at similar levels as several anthropogenic-impacted rivers and waters across China as well as aquaculture systems in Central Thailand, which are considered potential high-risk levels for human health. Aquaculture produces food for human consumption and aquaculture production will have impacts on downstream aquatic recipients of aquaculture pond water. This causes concern for the transfer of antibiotic resistance from aquaculture ponds to the general human population. Hence, there is an urgent need for improved management and monitoring of water quality in Bangladeshi aquaculture systems in order to mitigate the global antibiotic resistance crisis.

CRediT authorship contribution statement

Simon Bo Lassen: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Md. Emranul Ahsan:** Conceptualization, Methodology, Formal analysis, Writing – review & editing. **Seikh Razibul Islam:** Investigation. **Xin-Yuan Zhou:** Investigation. **Muhammad Abdur Razzak:** Supervision. **Jian-Qiang Su:** Writing – review & editing, Supervision. **Kristian Koefoed Brandt:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2021.151915>.

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